correspond to different kinds of particles. The key realization is that the detailed pattern of vibration executed by a string produces a specific mass, a specific electric charge, a specific spin, and so on—the specific list of properties, that is, which distinguish one kind of particle from another. A string vibrating in one particular pattern might have the properties of an electron, while a string vibrating in a different pattern might have the properties of an up-quark, a down-quark, or any of the other particle species in Table 12.1**.** It is not that an "electron string" makes up an electron, or an "up-quark string" makes up an up-quark, or a "down-quark string" makes up a down-quark. Instead, the *single* species of string can account for a great variety of particles because the string can execute a great variety of vibrational patterns.

Family 1		Family 2		Family 3	
Particle	**Mass**	**Particle**	**Mass**	**Particle**	**Mass**
Electron	.00054	Muon	.11	Tau	1.9
Electron-neutrino	$<10^{-9}$	Muon-neutrino	$<10^{-4}$	Tau-neutrino	$<10^{-3}$
Up-quark	.0047	Charm-quark	1.6	Top-quark	189
Down-quark	.0074	Strange-quark	.16	Bottom-quark	5.2

Table 12.1 The three families of fundamental particles and their masses (in multiples of the proton mass). The values of the neutrino masses are known to be nonzero but their exact values have so far eluded experimental determination.

As you can see, this represents a potentially giant step toward unification. If string theory is correct, the head-spinning, eye-glazing list of particles in Table 12.1 manifests the vibrational repertoire of a single basic ingredient. Metaphorically, the different notes that can be played by a single species of string would account for all of the different particles that have been detected. At the ultramicroscopic level, the universe would be akin to a string symphony vibrating matter into existence.

This is a delightfully elegant framework for explaining the particles in Table 12.1, yet string theory's proposed unification goes even further. In Chapter 9 and in our discussion above, we discussed how the forces of